

held-out LDS (gemma-3-1b, unseen scale)

Adam

Muon

LoRA-Adam

SGD

mean

0.8

0.7

0.6

0.5

0.4

0.3

0.2

0.1

0.0

0.39

0.70

0.37

0.66

0.34

0.63

0.28

0.58

0.35

0.64



this run's winner (solid)



prior best operator (hatched)